We have been given the docs Initial_Exploration and EER_diagram. Initial_Exploration contains the initial idea for us to create a database system for a website. It defines the weak/steong entities, relationships etc. EER_diagram, looks at Initial_Exploration and corrects some mistakes and creates the EER diagram.

Now we wish to create the relational model based on this EER diagram. Read the given ch9 to get to know how to convert EER diagram to relational model. Also refer to this summary:

1. **Strong entity types** mapped to relations with all simple attributes (and the simple components of composite attributes). Derived attributes need not be stored; if you do not store a derived attribute, say so in the explanation.
2. **Weak entity types** mapped to relations that include the primary key of the owner relation as a foreign key. The primary key is the owner's key combined with the partial key.
3. **Binary 1:1, 1:N, and M:N relationships** mapped using foreign keys or relationship relations, as appropriate. Relationship attributes must be placed correctly.
4. **Multi-valued attributes** mapped using one of the two options below. Choose the option according to the complexity of the attribute.
   (a) **Same relation:** If the attribute has a small, fixed maximum number of values that is known from the requirements (for example, at most two phone numbers), store it as that many separate columns in the same relation (for example, `phone_1`, `phone_2`).
   (b) **Separate relation:** If the number of values is not fixed or can be large, or the attribute has its own sub-attributes, create a separate relation. Its primary key combines the owner's primary key (as a foreign key) with the attribute value.

   For every multi-valued attribute, state in your explanation which option you chose, the maximum number of values (for option (a)), and a one-line reason.

   Note: A composite attribute, such as a name made of first name, middle name, and last name, is not multi-valued. As in item 1, it simply becomes separate columns in the relation of its entity type.
5. **Relationships of degree $n > 2$** mapped to a relation containing foreign keys to all participating relations.
6. **Subclasses and superclasses** (specialization or generalization) mapped using one of the textbook options. State which option you used.
7. **Recursive relationships** (if any) mapped with a foreign key to the same relation.
8. **Union types (categories)** (if any) mapped using the textbook option: create a relation for the category and, if its superclasses have different keys, give it a surrogate key that each superclass relation references as a foreign key. Any other EER construct not listed here must also be mapped using the corresponding textbook option; state which option you used.

Dont do any normalization. If you find and errors or gaps in the EER diagram correct and mention it a separate pdf report. Keep such changes to the minimal. In the pdf report, give a

short explanation of everything you have done, like which entity types, relationships, attributes became which relations and which options you chose. Keep explanation short and specific to our relations. Draw the actual relational diagram in a draw.io link.

Be careful to mention the following

- every relation with its name and all attributes;
- the primary key of each relation (for example, underlined);
- every foreign key, with an arrow from the foreign key to the primary key it references;
- enough size and clarity that all text is readable.